for loop :-

for (initialisation ; condition ; updation)

Swapping Two numbers :-

$a = 10, b = 20 \xrightarrow[\text{swap}]{\text{after}} a = 20, b = 10$

Ist way

int temp = a ;

a = b ;

b = temp ;

a $\boxed{\cancel{10}}$ 20

b $\boxed{\cancel{20}}$ 10

temp $\boxed{10}$

IInd way

a = a + b

b = a - b

a = a - b

a $\boxed{\cancel{10}}$ 30 (a + b) 20

b $\boxed{\cancel{20}}$ 10

b = a - b

= a + ~~b~~ - ~~b~~ = a

$a = a * b;$

$b = a / b;$

$a = a / b;$

$[-2^{31}, 2^{31} - 1]$

$(36 = 1, 2, 3, 4, 6, 9, 12, 18, 36)$

while $(i <= n)$

1×36

2×18

3×12

4×9

6×6

9×4 X

12×3 X

18×2 X

36×1 X

$36/6$

$\frac{36}{2}$

$\frac{36}{3}$

$\frac{36}{4}$

int n=48;

while(ixi<=n){

if(n%i==0){

$\rightarrow$ 6 $\leftarrow$ print(i);

6 $\leftarrow$ print(n/i);] X if (n/i != i)

(8 != 8)

i++;

}

}

48 $\Rightarrow$

1 X 48 $\rightarrow$ n/i

2 X 24 $\rightarrow$ n/i

3 X 16 $\rightarrow$ n/i

4 X 12 $\rightarrow$ n/i

6 X 8

8 X 6

12 X 4

16 X 3

24 X 2

48 X 1

64 $\rightarrow$ $\rightarrow$ 1 X 64

2 X 32

4 X 16

8 X 8

16 X 4

32 X 2

(64 X 1)

1, 64, 2, 32, 4, 16,

8,

ixi = ~~1~~ ~~4~~ ~~16~~, 64

i=8

H.W // Print Prime no with T.C.

Scanned with OKEN Scanner

```
for(int i=1; i<=100; i++) {
    print(i);
}
```

```
if(i==10 || i==20 || i==30 || i==40 || i==50)
{
    continue;
}
```

continue;

(10, 20, 30, 40, 50)

```
if(i==2) continue;
```

Scanned with OKEN Scanner

if ( )

continue; or $\underbrace{\text{break}}$;

We will skip that element

We will skip the whole iteration (i.e. We will move out from the loop).

```
for(i=1 -> 100) {
  if(i==1)
    break;
  sout(i)
}
```

break

1

100

2 ... 99

```
for(int i=1; i<=100; i++){
    if(i>=2 && i<=99)
         T        T
        continue;
    Sout(i);
}
```

11

100

1
2
3
4
5
.
.
.
.
100

$2^{31}-1$

Integer.MAX_VALUE

Integer.MIN_VALUE

-2^{31}

(534236469)

964632435 > $(2147483647)/10$

$2^{31}-1$

$n = 4444$

0

$2^{31}-1$

$\boxed{964632430}$

```
while (n != 0) {
    int digit = n % 10;
    revNo = (revNo x 10) + digit;
    n = n / 10;
}
return revNo;
}
```

964632430 > $\boxed{2147836/}$

-2^{31} 0 2^{31}

$(2^{31}-10) \times 10$

$(revNo > INT\_MAX/10)$

if (revNo > Integer.MAX_VALUE/10

|| revNo < Integer.MIN_VALUE/10)

$\times 10$

-2^{31} —— 0 —— $2^{31}-1$

H.W.

(-2147483412)

Integer.MAX_VALUE/10

Integer.MIN_VALUE/10

```
if (revNo > Integer.MAX_VALUE/10
    || revNo < Integer.MIN_VALUE/10)
    return 0;
}
```

Combination

A, B, C

0 1 2 3 4 5

$0 \to 9$

5^{10}

password of 5 digit →

(Permutation)

$6 \times 6 \times 6 \times 6 \times 6 = 6^5$

0 – 9

10 9 8 $= 10 \times 9 \times 8$

720 ways

1
2 ✓
3
.
.
.
9
0

A, B, C

A, B

B, C

A, C

$11C_{10} = 11$

$= \frac{11!}{10!(11-10)!}$

$25C_{10}$

(C, A)

12

21

2

A, B

B, A

1

$$\boxed{nC_r = \frac{n!}{r!(n-r)!}}$$

$10 \times 9 \times 8 \times 7 \times 6 \times 5 \times 4 \times 3 \times 2 \times 1$

$${}^{4}C_{2} = \frac{4!}{2!(4-2)!} = \frac{4!}{2! \times 2!}$$

$$= \frac{\cancel{4}^{2} \times 3 \times \cancel{2!}}{\cancel{2}! \times \cancel{2!}} = 6$$

picking 2 person from a group of 4 person.

$$\boxed{{}^{n}C_{r} = \frac{n!}{r!(n-r)!}}$$

A B C D

2 from 4

AB –

AC –

AD –

BC –

BD –

CD –

$(a+b)^2 = a^2 + 2ab + b^2$

$(a+b)^3 = a^3 + 3a^2b + 3ab^2 + b^3$

$(a+b)^4 = a^4 + 4a^3b + 6a^2b^2 + 4ab^3 + b^4$

$(a+b)^5 = a^5 + 5a^4b + 10a^3b^2 + 10a^2b^3 + 5ab^4 + b^5$

${}^5C_0 \quad {}^5C_4 \quad {}^5C_2 \quad {}^5C_1 \quad {}^5C_4 \quad {}^5C_5$

$\dfrac{5!}{0! \times (5-0)!} = \dfrac{\cancel{5!}}{1 \times \cancel{5!}} = 1$

$0! = 1$

${}^5C_2 = \dfrac{5!}{2! \times 3!} = \dfrac{5 \times 4 \times \cancel{3!}}{2! \times \cancel{3!}} = 10$

${}^5C_1 \quad 5 \times 4$

${}^nC_0 \quad {}^nC_n$

1

${}^nC_1$ or ${}^nC_{n-1}$

n

$(a+b)^6 = 1a^6 + 6a^5b^1 + 15a^4b^2 + 20a^3b^3 + 15a^2b^4 + 6a^1b^5 + 1b^6$

${}^6C_2$

${}^6C_3$

H.W:

$\frac{6!}{4!\,2!}$

$\boxed{(a+b)^{10}}$

$1!$

$4!\,2!$

Scanned with OKEN Scanner